

Closed Loop Control with Jump Processes Applied to OPEC Oil Production

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Abstract

OPEC is comprised of 13 oil-producing nations that coordinate production to influence global oil prices. The group must continually adjust output in response to changing market conditions and uncertain demand. To study optimal production behavior in this setting, a stochastic closed-loop control model is developed in which oil demand is subject to continuous volatility and rare but severe shocks, modeled via a Merton jump process. The associated Hamilton–Jacobi–Bellman partial integro-differential equation (HJB-PIDE) is derived and numerically solved using parameters calibrated from historical data. Sensitivity analysis demonstrates that small changes in price elasticity can substantially alter optimal production behavior. Despite this sensitivity a consistent optimal strategy pattern is found to be oscillatory production, where producers strategically underproduce to generate slight shortages, resulting in price hikes, which in turn increases future revenues. Finally, a comparison between bounded and unbounded price-response laws reveals that policy aimed at constraining price response fundamentally alters the structure of the optimal control problem and promotes more stable production outcomes.

Keywords

Production Modeling, Optimal Control, Jump Processes, Decision-Making Under Uncertainty, Energy Markets

1. Introduction

Oil production plays a significant role in the global economy through its influence on inflation, industrial output, financial markets, and geopolitical stability [1]. At each point in time, oil producers must determine an optimal oil production strategy while simultaneously balancing these effects along with future price shifts, demand responses, and the risk of large exogenous shocks to the financial market. This task is commonly referred to as the optimal production problem. Classical results from optimal commodity production have shown that stochastic uncertainty in price and demand fundamentally alter production strategies and introduces uncertain feedback loops that make producing time and rate decisions highly sensitive to volatility and future market conditions. As a result finding optimal oil production policies are hardly recovered by maximizing short term profits, but are instead found by solving dynamic control problems that consider long-run revenue, market stability, and resource allocation under uncertainty.

Much of the existing literature on optimal production of commodities adopt dynamic control frameworks where the producer's actions have little to no affect on the economic environment [2–4]. In such models, producers respond to their internal states but do not alter the underlying stochastic dynamics governing price or demand. In many settings, where a producer is sufficiently small with respect to the total level of production, such assumptions are justified. However unlike many commodities, oil production is highly concentrated among a small number of oil producing nations. Most notably, members of the Organization of Petroleum Exporting Countries (OPEC) collectively control 35%–40% of the world's oil production [5]. The members of the OPEC oil cartel often coordinate their oil production strategies and as a result can meaningfully influence market prices and demand patterns. These features motivate the use of closed-loop control models in which production decisions feed back into future price and demand dynamics. This feedback is especially important given oil's sensitivity to rare but severe demand shocks, such as the roughly 9% decline in global petroleum consumption during the COVID-19 pandemic, which culminated in negative West Texas Intermediate (WTI) oil futures prices [6].

To study optimal production in this setting, this paper develops a stochastic closed-loop control framework for OPEC oil production under demand uncertainty and market shocks. Demand is modeled using a jump diffusion process, and the associated Hamilton–Jacobi–Bellman partial integro-differential equation (HJB-PIDE) is derived and solved numerically using historically calibrated parameters. Such analysis reveals that optimal production policies generally

follow an underproduce–overproduce structure wherein producers create temporary shortages to raise prices then expand output to exploit those elevated prices. Next, the functional form governing price response to supply–demand imbalance fundamentally alters the structure of the optimal solution, where unbounded price responses incentivize aggressive boom–bust behavior while bounded response laws yield stable production schedules and dampen price volatility. These findings suggest that price-stabilization mechanisms can materially influence the production strategies that large oil producers optimally employ.

2. Approach

In this section structural assumptions of the model are detailed, then the stochastic dynamics governing oil demand and prices are developed with particular attention to how price responds to demand and how shocks are incorporated into demand dynamics. Finally the producer’s optimal production problem is formulated and the corresponding Hamilton-Jacobi-Bellman partial integro differential equation is derived.

2.1 Structural Assumptions

OPEC is comprised of 13 oil producing nations that coordinate their efforts to manipulate global oil prices through increasing or decreasing production. The group must constantly contend with changing market and demand conditions to adjust production accordingly. Sometimes member nations fail to meet their production quotas, while other times member nations intentionally produce and export more oil due to localized economic and financial pressure. The interplay between member countries along with its intrinsic game-like dynamics are an interesting area of research and have been explored in previous literature [7, 8]. Instead the model developed in this paper aims to describe an idealized case. First it is assumed that OPEC nations are able to coordinate and adjust their strategy instantaneously to respond to market conditions. Second, that all member countries are able to meet their production quotas or production cuts. Third, that if an OPEC nation fails to meet their obligations other member countries will not explicitly retaliate and will instead produce the remainder of the oil necessary to reach the quota. The first three assumptions allow for OPEC producers to be modeled as a single entity. Fourth, that oil is priced uniformly across all markets that OPEC operates. Fifth, it is assumed that any oil produced will be immediately purchased and consumed so long as there is demand for that oil and that the oil producers have no storage capacity.

2.2 Price & Demand Dynamics

Under the assumptions specified above, the price and demand dynamics are defined. The objective of such dynamics is not to recover the full structural model of price formation but instead to enforce a reduced form representation that captures three primary features of oil markets. First that prices respond asymmetrically to supply-demand imbalances. Second that demand is price sensitive and subject to rare but severe shocks. Third that both price and demand exhibit continuous volatility driven by external market forces. Let the price of oil at time t be P_t , the demand is denoted as D_t , and the total barrels of oil being produced at time t as q_t . The evolution of oil prices is then modeled as

$$dP_t = P_t[\eta \log\left(\frac{D_t + \epsilon}{q_t + \epsilon}\right)dt + \sigma_P dW_t^P], \quad (1)$$

where η represents the sensitivity of price to supply-demand imbalances. The logarithmic term is chosen to encode a relative scarcity mechanism, where if production exceeds demand the drift in price is negative while shortages generate positive price pressure. Importantly, unlike absolute difference terms, the log ratio captures the idea that the effect of shortages is dependent on the size of the market. The parameter ϵ is set to some small value to ensure the dynamics are well-defined. Note that $\sigma_P W_t^P$ is a Brownian motion corresponding to any small scale fluctuations in price caused by market pressures that are outside the producers control. Next, the demand evolves as a Merton jump diffusion process [9] with the dynamics

$$dD_t = D_t[-(\alpha_D - \beta P_t - \lambda k)dt + \sigma_D dW_t^D + (e^Y - 1)dN_t]. \quad (2)$$

Note that the dynamics encode that D_t is driven down when P_t rises and grows when P_t declines by some rate β . It is also assumed that a base level of demand growth α_D exists, which encodes industrialization trends and population growth. Given that demand suffers non-periodic shocks of various sizes, dN_t is incorporated into the dynamics, which represents the number of shocks within time dt drawn from a Poisson distribution of rate λ . In this setting $Y \sim \mathcal{N}(\mu, \delta^2)$ is the log-size of each jump. Note that the demand immediately before time t (i.e D_{t-}) is used, since a jump at t would result in a discontinuous process. Let $\sigma_D dW_t^D$ be small scale fluctuations in demand caused by market pressures external to the system. Finally let $k = \mathbb{E}[e^Y - 1]$, which prevents the dynamics from double counting shocks in both the drift and dN_t term.

The relevant probability space of the dynamical system described above is $(\Omega, \mathbb{F}, (\mathcal{F}_t)_{t=0}^T, \mathbb{P})$, where $\mathcal{F}_t$ is right continuous filtration generated by (W_t^D, W_t^P, N_t, Y_t) and $\mathbb{P}$ is the joint distribution of all the Brownian, Normal, and Poisson distributions.

2.3 OPEC Optimal Oil Production Problem

Next the producer's objective is defined. Let $V(D, P)$ denote the value function, representing the maximum discounted profit attainable by terminal time T if the producer starts with demand D and price P . In this setting the producer wishes to find a production policy $\{q_t\}_{t=0}^T$, where $q_t > 0$ for all t , that maximizes the expected discounted revenue over the time horizon $[0, T]$ with discount rate $\rho > 0$.

Note that $\min\{q_t, D_t\}P_t$ is the revenue at time t given the previously specified assumption that oil is immediately sold after production if there is sufficient demand. Production in excess of demand generates no revenue. The cost of production is modeled as the function $C(q_t) = c_1 q_t + c_2 q_t^2$ to de-incentivize endless production. The parameter c_1 is interpreted as the baseline cost of production per barrel of oil while c_2 is the marginal penalty for scaling production beyond current operational capacity. Finally note that the terminal reward is 0, meaning that all profits must be accumulated during the time horizon $[0, T]$. However the terminal condition can just as easily be some function $\psi(D_T, P_T, T)$ which penalizes unmet demand, or any number penalization or reward schemes. This formulation yields the producers optimization problem which is subject to the price and demand dynamics, (10) and (2) respectively,

$$V(D, P, t) = \sup_q \mathbb{E}^{D, P} \left[\int_0^T e^{\rho(t-T)} (\min\{q_t, D_t\}P_t - C(q_t)) dt \right]. \quad (3)$$

2.4 HJB-PIDE Derivation

Next the Hamilton-Jacobi-Bellman partial integro differential equation corresponding to the closed-loop control problem (3) is derived. To do so, dV_t is found by applying Itô's formula for jump processes to $V(D_t, P_t, t)$ [10]. For the purpose of brevity the value function evaluated immediately before a jump is denoted as $V^- = V(D_{t-}, P_{t-}, t)$, and let $d\tilde{D}_t = D_t - [(\alpha_D - \beta P_t - \lambda k)dt + \sigma_D dW_t^D]$ collects the continuous component of the demand dynamics. Then Itô's formula yields

$$\begin{aligned} dV_t = & \partial_t V^- dt + \partial_D V^- dD_t + \partial_P V^- dP_t + \frac{1}{2} [\partial_{DD}^2 V^- (d\tilde{D}_t)^2 + \partial_{PP}^2 V^- (dP_t)^2] \\ & + \partial_{DP}^2 V^- d\tilde{D}_t dP_t + [V(D_{t-} e^Y, P_t, t) - V(D_{t-}, P_t, t) - (e^Y - 1)D_{t-} \partial_D V^-] dN_t. \end{aligned} \quad (4)$$

Note that the formula for dV intuitively captures the change in V with respect to each input (t, D_t, P_t) , the change due to Brownian shocks, along with the change due to jumps beyond first order effects. Next the associated HJB-PIDE is derived using the dynamic programming principle and the infinitesimal generator dV_t . For a small time increment $h > 0$, the value function satisfies

$$\begin{aligned} V(D, P, t) = & \sup_{q_t} \mathbb{E}^\mathbb{P} \left[\int_t^{t+h} e^{-\rho(s-T)} (\min(q_s, D_s)P_s - C(q_s)) ds + e^{-\rho h} V(D_{t+h}, P_{t+h}, t+h) \right] \\ \xrightarrow{h \rightarrow 0} 0 = & \sup_{q_t} \left\{ \mathbb{E}^\mathbb{P} \left[e^{-\rho(t-T)} (\min(q_t, D_t)P_t - C(q_t)) dt \right] + \mathbb{E}^\mathbb{P} [dV_t] \right\}. \end{aligned} \quad (5)$$

To evaluate (5), expand the expectation of the infinitesimal increment dV_t and use the standard identities $\mathbb{E}^\mathbb{P}[dW_t^{D/P}] = 0$, $\mathbb{E}^\mathbb{P}[dN_t] = \lambda dt$, $(dW_t)^2 \sim dt$. The general computations are omitted as they are standard. Using the expanded form of $\mathbb{E}^\mathbb{P}[dV_t]$ and letting ϕ be probability density function of Y , the HJB-PIDE is recovered as

$$\begin{aligned} 0 = & \partial_t V + \partial_D V D_t (\alpha_D - \beta P_t) + \frac{1}{2} \partial_{DD}^2 V D_t^2 \sigma_D^2 + \frac{1}{2} \partial_{PP}^2 V P_t^2 \sigma_P^2 + \partial_{DP}^2 V D_t P_t \sigma_D \sigma_P \rho \\ & + \lambda \left(\int_{-\infty}^{\infty} V(D_t e^y, P_t, t) \phi(y) dy - V(D_t, P_t, t) - \partial_D V D_t k \right) \\ & \sup_{q_t} \left\{ e^{-\rho(t-T)} (\min(q_t, D_t)P_t - C(q)) + \partial_P V P_t \eta \log \left(\frac{D_t + \epsilon}{q_t + \epsilon} \right) \right\}, \end{aligned} \quad (6)$$

subject to $V(D_T, P_T, T) = 0$. Note that the HJB-PIDE is deterministic, so the dependence on D_{t-} can be dropped.

3. Results

Next a numerical simulation scheme is applied to solve for the optimal revenue $V(D_0, P_0)$ and the optimal production strategy q^* within the time horizon $[0, T]$. First appropriate model parameters are found in existing econometrics literature, then a Monte Carlo scheme is implemented to solve the HJB-PIDE in equation (6). This section concludes with a qualitative analysis of the optimal oil policies along with a critical analysis of price-demand response dynamics.

3.1 Setting Parameters

Recall that the model has 10 parameters, $\{\eta, \sigma_P, \alpha_D, \beta, \lambda, \sigma_D, \mu, \delta, c_1, c_2\}$. Each parameter is calibrated from empirical econometric studies. First, recall that η represents the sensitivity of oil price to shortages and surpluses, otherwise referred to as the short-run elasticity of oil demand which is estimated to be anywhere between 5% to 10% annually. In other words a 1% imbalance in demand vs. supply can induce a 5% to 10% price change [11]. Consequently η is set to 1.075. The parameter σ_P represents the volatility of oil prices which on average is estimated to be approximately 0.25 annually. [11].

The first parameter of the demand dynamics is α_D which represents the baseline demand growth from populations growth and industrialization. Empirical estimates indicate that oil consumption grows by about 1.5% annually [12]. Next is β , which represents the price effect on demand. Qualitative analysis indicates that only for large fluctuations in price is demand dampened. So β is estimated to be a small value of 0.01 [11]. Finally σ_D represents the volatility in demand, which is estimated to be about 1% annually [11].

The jump process has three parameters. First λ is set to 0.1 annually, which represents a major demand disruption every 10 years [12]. The mean log jump size μ is set to -0.05 and the standard deviation $\delta = 0.1$ which corresponds to an average $e^Y - 1 \approx -5\%$ plunge in demand. These parameters are inline with recent demand shocks which almost always skew negative, with the largest negative and positive shocks in the last 10 years being -9% in 2020 and 3.3% in 2010 [12].

The production cost function $C(q)$ is quadratic with two parameters c_1 and c_2 . The parameter c_1 represents the baseline production cost per barrel which is estimated to be about \$10, and the marginal cost c_2 is estimated to be about \$0.1 [13]. Finally ρ is treated as a measure of inflation and is set to the federal reserve's annual inflation target of 2%.

3.2 Monte Carlo Solver

The HJB-PIDE for the optimal production strategy q_t does not permit a closed form solution and as a result is solved using a Monte Carlo neural network approach. The production policy q_t is parameterized as a trainable neural network $\pi_\theta(D_t, P_t, t)$ with parameters θ . For the purpose of this analysis a fully connected multi-layer-perceptron is employed with three hidden layers of input dimension 256, 128, and 64 using Pytorch 2.9.1 on a 16 core Linux machine with 16GB of RAM. Additionally the Euler-Maruyama scheme is employed with one month intervals, which yields the forward recursive update rules

$$P_{t+\Delta t} = P_t \left(1 + \eta \log \left(\frac{D_t + \epsilon}{q_{i,j}^{n*} + \epsilon} \right) \Delta t + \sigma_P \sqrt{\Delta t} \xi_P \right), \quad (7)$$

$$D_{t+\Delta t} = D_t \left(1 + (\alpha_D - \beta P_t - \lambda k) \Delta t + \sigma_D \sqrt{\Delta t} \xi_D + (e^Y - 1) dN_t \right), \quad (8)$$

where $\xi_P, \xi_D \sim \mathcal{N}(0, 1)$ are independent standard normals, $Y \sim \mathcal{N}(\mu, \delta^2)$, and dN_t is a Bernoulli random variable with success probability $\lambda \Delta t$. Training proceeds over a fixed number of epochs. At each epoch, a collection of M independent sample paths $\{D_t^j, P_t^j\}_{j=1}^M$ are generated using the Euler-Maruyama scheme and the parameters θ of the neural network π_θ are tuned to minimize

$$\mathcal{J}(\theta) = -\mathbb{E} \left[\sum_{j=1}^M \sum_{n=0}^{N-1} r_{t_n}^j \Delta t \right], \quad r_t^j = e^{-\rho(t-T)} \left(\min(q_t, D_t^j) P_t^j - C(\pi_\theta(t, D_t^j, P_t^j)) \right). \quad (9)$$

By repeatedly alternating between Monte Carlo simulation and policy updates, the procedure converges to a locally optimal production strategy that maximizes expected discounted revenue under the modeled price and demand dynamics.

3.3 Optimal Policies

The Monte Carlo algorithm is run for 100 epochs for several parameter configurations and price-demand laws. First the optimal production strategy under the logarithmic price-demand response law specified in equation (10) are analyzed, which encodes a proportional scarcity mechanism with unbounded price drift. Solving the HJB-PIDE using the Monte Carlo algorithm, the optimal policy is found to exhibit aggressive manipulation of supply. In particular the producer initially suppresses production to generate shortages which induce rapid price increases. This price spike is then exploited through an abrupt increase in production which results in large but short term revenues. The resulting dynamics are presented in figure (1a). While such strategies are optimal they lead to rapid collapse in demand as price rises to an untenable level. Such behavior occurs when prices rise sharply and remain elevated, resulting in the demand drift term $(\alpha_D - \beta P_t)$ remaining negative. This boom-bust behavior can be interpreted as a loss of market trust which may result in a structural market de-coupling from oil.

Adjusting the baseline demand and sensitivity parameters to $\alpha_D = 0.6$ and $\beta = 0.001$ yields a qualitatively similar but more sustainable oil producing strategy. The resulting dynamics are presented in figure (1b). In this regime the producer periodically creates shortages in order to induce a jump in price which the producer then exploits by increasing production. However as a result of the increase in production, a surplus is created which decreases the price. The producer then responds to the decrease in price by creating a shortage. This cyclical pattern of behavior, while volatile, proves to be highly profitable for the producer in a setting where stochastic shocks may dramatically decrease profitability at random points in time.

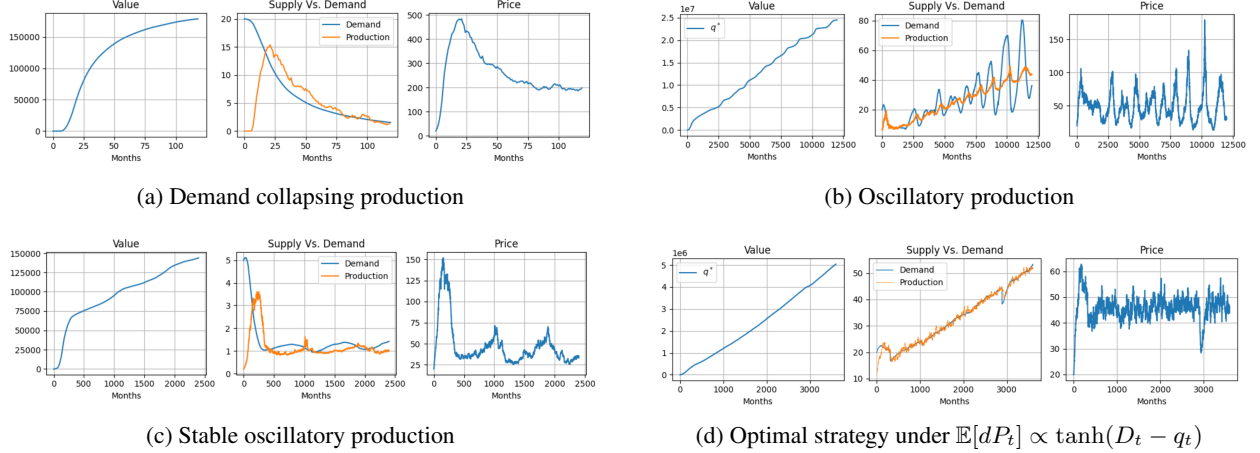


Figure 1: Optimal production under alternative price-demand response dynamics and parameter configurations

The magnitude and frequency of these oscillations are highly sensitive to the elasticity parameter η . Larger values of η amplify the responsiveness of prices to supply-demand imbalances which increases the volatility of price to the producers underproduce-overproduce strategy. Conversely reducing η dampens the price responses and stabilizes the oscillatory production strategy without fundamentally changing the incentive to employ a underproduce-overproduce strategy. The optimal dynamics for a regime where $\eta = 1.01$ is presented in figure (1c).

Note that naïve production strategy is to always have production meet demand. Under such a policy expected price drift vanishes, price remains approximately constant at P_0 , demand continues to grow exponentially, and the quadratic production cost eventually dominates revenues. In such a setting revenues will eventually go to negative infinity over an infinite time-horizon. This behavior highlights that, under unbounded price-response laws, the optimal policy necessarily involves deliberate supply distortion when price feedback is endogenous. Specifically the general class of underproduce-overproduce (boom-bust) strategies are optimal across a variety of parameter configurations.

To understand whether a underproduce-overproduce strategy persists with a more conservative price law the alternative price dynamics

$$dP_t = P_t[\eta \tanh(D_t - q_t)dt + \sigma_P dW_t^P] \quad (10)$$

is introduced. The tanh response law preserves the same directional behavior, however unlike the log response law the

drift in this case is bounded from $[-\eta, \eta]$. This alteration forces the producer to engage in more conservative strategies since their ability to manipulate the market price is significantly diminished. Under this setting the optimal strategy is to slightly but consistently underproduce which is presented in figure (1d). In fact across $M = 10^4$ simulations with $T = 200$, $\frac{1}{T \cdot M} \sum_{i=1}^M \sum_{t=1}^T (D_t - q_t) = 0.4438$. Importantly, this stabilization is achieved with little loss in cumulative value relative to the unbounded case and highlights the impact of price-response policy on fundamentally changing the optimal production strategy for OPEC aligned oil producers.

4. Conclusion & Final Thoughts

Operationally, our results indicate that oil producers with sufficient market share, like OPEC, should prioritize price discipline over short term volume maximization in the presence of demand shocks. The optimal policies identified consistently favor underproduction and controlled price exploitation, instead of aggressively meeting demand. Additionally the analysis highlights that production strategies that induce extreme price spikes risk triggering sustained demand contractions through a loss of customer confidence, policy backlash, or the adoption of oil substitutes. Finally the comparison between unbounded and bounded price response dynamics is particularly informative. While unbounded price response incentivizes oil production that aggressively manipulates price and results in boom-bust cycles, bounded price response laws create a regime where boom-bust strategies are sub-optimal. Practically, this insight implies that institutional or policy mechanisms which implicitly constrain price responses—such as production smoothing rules, quota adjustment bands, or coordinated response frameworks—may help stabilize oil markets by discouraging destabilizing boom-bust production behavior.

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